

THE 4-STAGE QUALIFICATION LADDER & CAUSAL-LOOP VERIFICATION MATRIX

Systematic Progression Across the Throughput-Fidelity Continuum: Software Simulation to In-Situ Physical Fault Injection

THROUGHPUT / SAMPLE VOLUME:

10⁶ runs/day (Massive Parallel)

10⁴ runs/day

10² runs/day

10⁰–10¹ runs/day (Constrained) ➔

HARDWARE & PHYSICAL FIDELITY:

0% (Idealized Math / ODEs)

40% (Target Silicon)

85% (Real Dyno Load)

100% (Full Physical Reality)

STAGE 1 · Software-in-the-Loop

Multi-Physics Simulation (SIL)

★ Statistical Parameter Sweeps

TEST SUBSTRATE:

Cloud GPUs / MuJoCo / Isaac Sim

CAUSAL-LOOP FIDELITY:

- **Algorithm / Policy:**
Real Python / C++ binary
- **Target Silicon / Clocks:**
Simulated / Host CPU time
- **Bus Transport & Latency:**
Zero-latency idealized copy
- **Actuation & Dynamics:**
Rigid-body ODE approximations
- **Environmental Physics:**
Synthetic contact models

✓ WHAT IT ESTABLISHES:

Broad combinatorial state-space coverage, policy loss convergence, nominal trajectory tracking.

× EVIDENTIARY BLIND SPOT:

Silicon bus contention, clock skew, thermal derating, real fluid cavitation, unmodeled friction.

STAGE 2 · Processor-in-the-Loop

Target SoC / MCU Emulation (PIL)

★ Silicon Timing & Interrupts

TEST SUBSTRATE:

Target ARM Cortex-A/M + FPGA Plant

CAUSAL-LOOP FIDELITY:

- **Algorithm / Policy:**
Real compiled firmware binary
- **Target Silicon / Clocks:**
Real SoC oscillator & registers
- **Bus Transport & Latency:**
Real AXI / SPI / CAN buffers
- **Actuation & Dynamics:**
Virtual PWM capture (FPGA)
- **Environmental Physics:**
Real-time 10 kHz math model

✓ WHAT IT ESTABLISHES:

Zero dynamic malloc compliance, interrupt latency tails, RTOS context-switch deadlines, watchdog resets.

× EVIDENTIARY BLIND SPOT:

Motor back-EMF, inverter power-rail droop, cable harness EMI, mechanical backlash.

STAGE 3 · Hardware-in-the-Loop

Actuator Dynamo Stand (HIL)

★ Electrical & Thermal Load

TEST SUBSTRATE:

Real Inverters, Motor & 4Q Dyno

CAUSAL-LOOP FIDELITY:

- **Algorithm / Policy:**
Real target firmware & weights
- **Target Silicon / Clocks:**
Real multi-rate heterogeneous SoC
- **Bus Transport & Latency:**
Real physical transceivers & harness
- **Actuation & Dynamics:**
Real BLDC motor & dynamic brake
- **Environmental Physics:**
Active dyno counter-torque (1 kHz)

✓ WHAT IT ESTABLISHES:

Inverter thermal dissipation at 10 A, motor back-EMF, regenerative braking torque, power-rail droop.

× EVIDENTIARY BLIND SPOT:

Hydrodynamic water hammer, fluid contamination, pipe vibration, acoustic standing waves.

STAGE 4 · Physical Fault Rig

In-Situ Fluid / Vehicle Stand

★ Destructive Physical Limits

TEST SUBSTRATE:

Full Assembled Machine in Blast Cell

CAUSAL-LOOP FIDELITY:

- **Algorithm / Policy:**
Production flashing on chassis
- **Target Silicon / Clocks:**
Full vehicle harness & power rail
- **Bus Transport & Latency:**
Full harness with ambient EMI
- **Actuation & Dynamics:**
Real motorized valves & mechanics
- **Environmental Physics:**
Real high-pressure fluid & wear

✓ WHAT IT ESTABLISHES:

Hydraulic water hammer, elastomeric seal extrusion, mechanical jamming, pipe burst containment.

× EVIDENTIARY BLIND SPOT:

Rare combinatorial parameter sweeps (destructive cost limits sample count to $N \leq 50$).